Supplementary material to AN APPROACH TO ESTIMATION IN RELATIVE SURVIVAL REGRESSION

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A: Technical details

Let the patients (indexed by i or j) be ordered in time ($t_i \geq t_{i-1}$), where t_i denotes the exit time for patient i. Denote the baseline excess hazard by λ_0, the cumulative baseline excess hazard by Λ_0, the population hazard for patient i by λ_{Pi} and the cumulative population hazard for patient i by Λ_{Pi}. Vector z_i represents the covariate values for the ith patient.

We use $Y := \{\delta, t, z\}$ to denote the observed data, and let $X := \{Y, \delta_E\}$ denote the complete data. The complete data likelihood can be written as

$$L(\Theta|X) = f(X|\Theta) = \prod_{i=1}^{n} \left(\lambda_0(t_i)e^{\beta' z_i}\right)^{\delta_{Ei}\delta_i} \lambda_{Pi}(t_i)^{(1-\delta_{Ei})\delta_i} e^{-\{\Lambda_0(t_i)exp(\beta' z_i)+\Lambda_P(t_i)\}} \quad (1)$$

where $\Theta = \{\beta, \lambda_0\}$. The observed likelihood is given by

$$L(\Theta|Y) = g(Y|\Theta) = \prod_{i=1}^{n} \left\{\lambda_0(t_i)e^{\beta' z_i} + \lambda_{Pi}(t_i)\right\}^{\delta_i} e^{-\{\Lambda_0(t_i)exp(\beta' z_i)+\Lambda_P(t_i)\}}.$$

Now define k as

$$\begin{aligned} k(\delta_E|Y,\Theta) := \frac{f(Y,\delta_E|\Theta)}{g(Y|\Theta)} &= \prod_{i=1}^{n} \left[\frac{\left\{\lambda_0(t_i)e^{\beta' z_i}\right\}^{\delta_{Ei}} \left\{\lambda_P(t_i)\right\}^{1-\delta_{Ei}}}{\lambda_0(t_i)e^{\beta' z_i} + \lambda_{Pi}(t_i)}\right]^{\delta_i} \\ &= \prod_{i=1}^{n} k_i(\delta_{Ei}|Y_i,\Theta) \ \ \text{(say)}, \end{aligned}$$

and note

$$k_i(\delta_{Ei}|Y_i,\Theta) = \begin{cases} \delta_i \frac{\lambda_{Ei}}{\lambda_{Oi}} & : \quad \delta_{Ei} = 1 \\ & : \\ \delta_i \frac{\lambda_{Pi}}{\lambda_{Oi}} & : \quad \delta_{Ei} = 0. \end{cases} \quad (2)$$

From Dempster et al. (1977), the equality on which the EM is based is

$$\log L(\Theta|Y) = \log L(\Theta|Y, \delta_E) - \log k(\delta_E|Y, \Theta). \tag{3}$$

Taking expectations with respect to (2) and noting that the left side does not depend on δ_E, equation (3) can be rewritten as

$$\log L(\Theta|Y) = E\left\{\log L(\Theta|Y, \delta_E)\right\} - E\left\{\log k(\delta_E|Y, \Theta)\right\}. \tag{4}$$

Differentiating (4) twice shows that the observed information can be calculated as the complete information minus the missing information.

To show that the Cox likelihood can be used in place of the complete likelihood (1), we show that we get the same result by profiling λ_0 out of the complete likelihood.

The part of the logarithm of the complete likelihood that depends on β and λ_0 is

$$\log f(X|\Theta) = \sum_{i=1}^{n} \left\{\delta_i \delta_{Ei} \left(\log \lambda_0(t_i) + \beta' z_i\right) - \Lambda_0(t_i) e^{\beta' z_i}\right\}. \tag{5}$$

As usual, the nonparametric maximum likelihood estimator of λ_0 is zero everywhere except at observed death times (Andersen et al., 1993, Chapter 4). Rewriting $\lambda_{0j} := \lambda_0(t_j)$ and

$$\Lambda_0(t_i) = \sum_{j=:t_j \le t_i} \lambda_{0j},$$

interchanging the order of summation and noting $\delta_i \delta_{Ei} = \delta_{Ei}$, we get

$$\log f(X|\Theta) = \sum_{i=1}^{n} \left\{\delta_{Ei}(\log \lambda_{0i} + \beta' z_i) - \lambda_{0i} \sum_{j \in R_i} e^{\beta' z_j}\right\}. \tag{6}$$

This leads to

$$\hat{\lambda}_{0i} = \frac{\delta_{Ei}}{\sum_{j \in R_i} e^{\beta' z_j}},$$

and substitution into (6) gives

$$\log f(X|\Theta) = \sum_{i=1}^{n} \left\{ \delta_{Ei} \left(\beta' z_i - \log \sum_{j \in R_i} e^{\beta' z_j} \right) + \delta_{Ei} \right\}.$$

The score obtained from this expression is therefore the same as obtained from the usual partial likelihood:

$$L(\Theta|X) \propto \prod_{i=1}^{n} \left(\frac{\exp(\beta z_i)}{\sum\limits_{j \in R_i} \exp(\beta z_j)} \right)^{\delta_{iE}}.$$

The problem of a non-decreasing $r(t)$ can be solved by using a kernel function smoothing procedure for estimating the baseline excess hazard (Ramlau-Hansen, 1983). Let $\lambda_0^{(k)}(t_i)$ denote the estimated baseline hazard at step k and $\lambda_0^{(k)}(t_i, b(t_i))$ its smoothed version:

$$\lambda_0^{(k)}(t_i, b(t_i)) = \frac{1}{b(t_i)} \int_0^{\infty} K\left(\frac{t_i - u}{b(t_i)}\right) \lambda_0^{(k)}(u) du,$$

where $K \geq 0$ is a left-continuous bounded kernel function having integral 1, support $[0, 1]$, and $b \geq 0$ denotes the window (band-width). The band-width can be allowed to change with time, as there are more individuals at the beginning of the follow-up and therefore more events, while the intervals between the events at the end of the follow-up tend to be longer. Special care must also be devoted to the so called boundary effects at the beginning of the follow-up, where the support for K exceeds the available range of the data. This can be done by allowing the shape of the kernels to change in the boundary, see for example Müller and Wang (1994).

We have chosen to work with the Epanechnikov kernel function (Silverman, 1986) on interval $[0, 1]$; i.e. $K = 1.5 * (1 - x^2)$. We allow the band-width to change four times during the follow-up interval, always proportional to the maximum time difference between two consecutive events. A the beginning of the interval, while the time elapsed is smaller than the band-width, we let the band-width change so that it is equal to the time. This option seems to give satistfying results. The beginning of the interval is anyway usually not problematic as the probability of excess hazard death tends to be much larger than the probability of a population reasons related death.

The factor of proportionality is chosen by the user through argument `bwin`, but it can also be estimated automatically by specifying `bwin=-1`. In this case, the program estimates the relative cumulative hazard, fits the model with no covariates with several band-widths, and chooses the band-width that gives the smallest difference between the two curves. It is important to note that only a small amount of smoothing is needed, as its only goal is to ensure the monotonicity of the cumulative baseline hazard function. An oversmoothed curve on the other hand already implies making assumptions about the baseline hazard function form.

B: Properties of the EM based approach

We performed a simulation study in order to evaluate the properties of the EM approach and compare it with the standard fully parametric approaches. The simulations were designed so that the model assumptions hold regardless of whether we use splines or the piecewise constant parametric model, i.e. the baseline excess hazard was simply taken to be constant in time. We let all the cases have the same values of demographic variables, corresponding to a 73 year old male with infarction in 1980, and used the appropriate population hazard taken from Slovenian mortality tables (λ_P starts at the value 0.08 per year). We included a single additional binary covariate Z, with $E[Z] = 0.5$, and took coefficient $\beta = 1$. A more complex example, where the assumptions of the parametric models do not hold, is given in Section 6 of the paper.

To illustrate how the proportion of disease-specific deaths influences the quality of coefficient estimation, we varied the size of the baseline excess hazard (λ_{E0} varies from 0.02 to 1.5 per year) to get different proportions of disease-specific deaths. The survival times due to each risk (λ_P and λ_E) were simulated separately and the minimum of the two survival times was taken. Additionally, information on the cause of death ($\mathcal{C}$ or other) was stored. In this way we also obtained the complete data set, to which the cause-specific Cox model could be fitted serving as a reference for all the comparisons. Each baseline excess hazard size was simulated 500 times with sample size $n = 1000$, without censoring. Three models were fitted to each simulated data set: the cause-specific Cox model (denoted as complete), the additive model with a nonparametric baseline excess hazard (EM), and a classical additive model with a correctly assumed constant baseline excess hazard.

The means of the coefficients estimated by each of the three methods and the coverage are presented in Figure 1a. The estimates are very close when the proportion of the excess hazard deaths is high but both relative survival model estimates deteriorate as fewer deaths are due to $\mathcal{C}$. In particular, both the parametric and the EM methods seem to be biased when the percentage of condition-attributable deaths falls under 30%, and as both methods also tend to run into convergence problems the additive model in these situations does not seem to be a good choice. The coverage is satisfactory in all cases and does not seem to exhibit any particular trends.

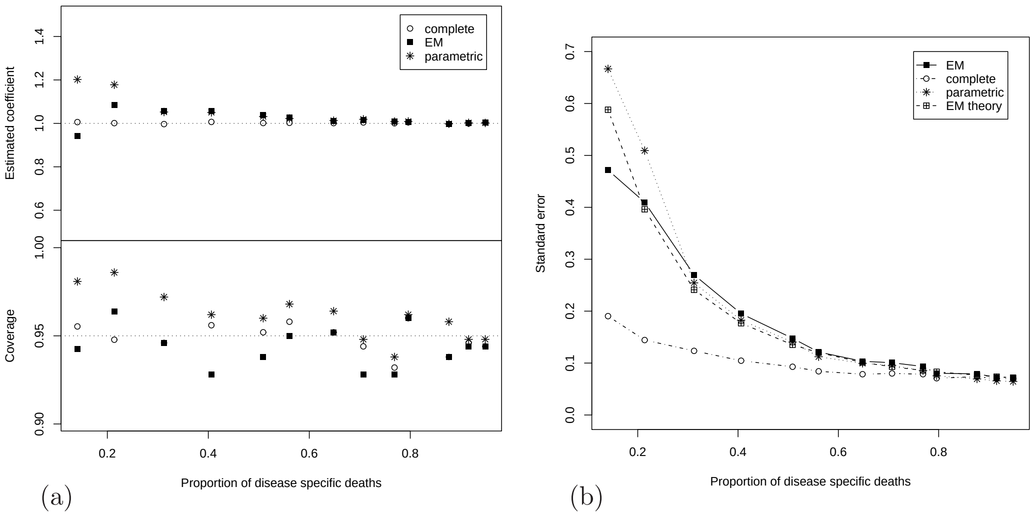

Figure 1: *The means, the coverage (a) and standard deviations (b) of the estimated coefficient under the EM approach (EM) compared with those under the parametric model (parametric) and Cox model with complete observation (complete). The average of the estimated standard errors under the EM approach (EM theory) is also given.*

Figure 1b explores the properties of the variance of the estimated coefficients. In each simulation we obtained the information-based variance estimate for the coefficient and the plot shows the square root of the average of these (denoted as EM theory) together with the empirical standard deviation of coefficient estimates across simulations (denoted as EM). The two values are close, implying that formula (6) of the paper provides a good estimate. The standard deviation of the coefficients estimated using maximum likelihood, with correctly assumed constant baseline excess (parametric), is generally close to the EM values. Both incomplete data methods perform worse than the complete data Cox method, as expected. The variance decreases for the latter method as the proportion of disease-specific deaths increases, because other deaths are treated as censoring and their proportion correspondingly decreases. The other methods have a similar pattern as the observed data variance is closely related to the complete data variance. When the proportion of disease-related deaths is very high, the precision of the incomplete data estimates is no worse than when cause of death information is known, for this example at least. We can conclude that the variance of the additive model strongly depends on the proportion of deaths due to excess risk. If the survival of the observed cohort is very similar to the survival of the population, the variance of the estimated coefficients for excess hazard can become very large, regardless of the fitting procedure.

Based on the simulations, we can conclude that the additive model may not be the best choice in situations with less than 30% deaths due to the excess risk, this coincides well with the recommendations of Sasieni (1996). Instead, one should rather turn to using the multiplicative model (Andersen et al., 1985) or the transformation approach (Stare et al., 2005).

As a final note, we study the effect of partially known causes of death on the variance of the parameter estimates. We simulate a data set and start by fitting a Cox model, i.e. by having the complete cause of death information. Cause of death was then taken to be unknown for a random sample of either 0%, 25%, 50%, or 75% of individuals, and the EM method that takes into account the known values applied (see Section 4 of the paper). The results are presented in Table 1 and we can clearly see how the variance increases as less information is available, albeit only modestly.

% known	$\hat{\beta}$	SE
100	1.011	0.174
75	1.011	0.179
50	1.010	0.183
25	1.009	0.185
0	1.007	0.188

Table 1: *Results of* 1000 *simulation runs depending on the percentage of known causes of death. The Cox model is fitted in the case of all the information being known (100%), the other four cases are fitted using the EM procedure.*

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